

Sustainability Tool Instructions

This document is an instruction on how to make an absolute sustainability assessment using the "Sustainability tool". It is meant as a companion to the tool, and should be consulted in case of confusion, rather than read from start to finish. For some explanations, examples will be provided referring to an example case study comparing the absolute sustainability of light bulbs to that of candles. To use the 'Sustainability tool', you need a functional unit and a list of materials and processes that are included in the cradle-to-grave life cycle of the products or processes studied.

1 Ecoinvent Database

In the sheet 'Ecoinvent Data', you can find a copy of the Ecoinvent database, with about 25 000 processes. For each process, several environmental indicators are given, as seen in Figure 1.

carbon footprint	remaining human health damage	remaining ecosystem quality damage	fossil and nuclear energy use	water scarcity footprint	global warming potential (GWP100)	remaining human health damage
kg CO2-Eq	Min. lost	PDF.m2.yr	MJ deprived	m3 world Eq deprived	kg CO2-Eq	DALYs

Figure 1: Ecoinvent indicators with corresponding units

2 Selected Processes

The 'selected processes' sheet is referring to the Ecoinvent database, and is intended to give you an overview of the most relevant processes for each calculation. The recommendation is to list desired processes in the first column, and then match with the appropriate process number from the Ecoinvent database. This should populate the remaining columns of the table with the data from the Ecoinvent database. As there might not be an exact match between desired and available data, this also allows you to keep an overview of the assumptions made.

3 Calculating the Carbon Footprint

Once you have identified the process numbers, these should be added to column B in the 'Carbon & health footprint' sheet of the tool. This should fill in the entire row, except for column E,

F, and G. Note that this function will only work if the process number exists in the 'selected processes' sheet. In column E, the total amount of this process needed for one unit should be added, and in column F the unit. In column G, the corresponding amount per FU should be added.

Once the process number and amount per FU have been added, the remaining columns I through U will be filled in, showing the impact of the specific process as a carbon footprint, on human health, ecosystem quality, and non-renewable primary energy resources. The final column ('Check [gCO₂e/MJ]'), can be used as a check, as an unrealistic value is probably due to an earlier error. For each indicator, the total impact of the system per FU can be found at the bottom of the column.

3.1 Avoided Heat or Electricity

In the tool, it is possible to account for avoided heat or electricity from incinerated waste. For this, you need the weight of the incinerated mix, as well as its energy content in MJ/kg. You should also add a process for heat or electricity to the disposal phase in your 'carbon calculator' sheet, where you will subtract the amount of avoided heat or electricity.

To find the amount of avoided heat, the energy content and weight should be multiplied with the efficiency factor for heat. The same method can be used to calculate the amount of avoided electricity, but the unit should also be converted from MJ to kWh.

3.2 Example

If we consider an example case of using candles to illuminate a room, the total amount of paraffin (10g) needed to produce one candle should be added to the 'Quantity per unit' column, i.e. column E. Then, column G, 'quantity per FU', should be updated depending on how many units are needed for 1 functional unit. In this case the functional unit 720 lumen provided for 1 hour, which equates to an average use of 15 candles per hour. Hence, the quantity per FU is 150g, or 0.15 kg. See Fig. 2. Note that different units are used in column F and H. In column F, any unit can be used, but in column H, the unit must match the data from the database.

	E	F	G	H
on	Quantity per unit	Unit	Quantity per FU	Unit
	Per unit produced		Per FU	
	10	g	0.15	kg

Figure 2: Example from the 'Carbon calculator' sheet.

Fig. 3 shows an example of how we can calculate the avoided electricity from incinerated waste. Firstly, we need a process for the handling and burning of the waste in the incineration plant.

=G35*I11*I5/I4			
F	G	H	I
nt tool			
Transformation multiplier		1000	t/kg, kg/g...
Conversion factor		3.6	MJ/kWh
Conversion efficiency - Electricity		25%	
Conversion efficiency - Heat		60%	
Material		Energy content	
Incineration mix - candle		40	MJ/kg
Per FU			
0.015		kg	0.04 kg CO2eq/l
=G35*I11*I5/I4		kWh	0.16 kg CO2eq/l
			kg CO2eq/l

Figure 3: Avoided electricity formula

For this, we need to know the weight of the incinerated waste, which in this case is 0.015 kg. We multiply this by the energy content (cell I11, red), the conversion efficiency (cell I5, purple), and divide by the conversion factor (cell I4, green) to convert from MJ to kWh. The same process should be followed to find the heat recovery, except that the conversion factor is not then necessary, and that the conversion efficiency of the electricity can be removed.

3.3 Expand the Sheet

Note that the 'carbon calculation' sheet is divided per life-cycle stage. This is to simplify debugging and analysis. However, the number of rows per life cycle stage given will probably **not** correspond to the number of processes for your case study. If there are unused rows, they can simply be removed. If you need more rows, you can freely insert rows into the sheet. When doing so, it is important to preserve the formulas in column C and D, as well as columns H-L. This can be done by simply pulling them down from the rows above, or copy-pasting them from an earlier row.

4 Health dimensions

Another aspect of sustainability that can be calculated using the excel sheet is contained in cell referenced to *Human health*. In *selected processes* tab the cell titled *min.lost* value per unit is already assigned from the corresponding process # taken from the ecoinvent data or needs to be filled out based on the literature (Fig. 11). While making calculations, the inputted process # will provide value of minutes lost, which then needs to be normalized to the

functional unit (FU). The result for analysis is *Total min lost* and *fraction human health* for identifying the most impactful stage of the life cycle (Fig. 12).

Process #	Ecoinvent Activity Name	Geography	Reference Product Name	Reference Product Unit	Reference Product Amount	Carbon footprint [kg CO2-Eq]	Human health damage [Min. lost]
	Materials						
14151	platinum to generic	GLO	metal catalyst for catalytic conv	kg	1	45758.650	131821.85
17333	market for polyethy	GLO	polyethylene, high density, gran	kg	1	2.835	0.93
17481	market for polyuret	RoW	polyurethane, rigid foam	kg	1	6.062	3.00
8639	market for ethylene	RER	ethylene	kg	1	1.823	0.27
21711	tap water production	RoW	tap water	kg	1	0.000	0.00
10352	heat production, na	Europe without Switzerla	heat, district or industrial, natur	MJ	1	0.075	0.00
16715	market for petrol, lo	Europe without Switzerla	petrol, low-sulfur	kg	1	1.037	0.31

Figure 11: Human health lost cell location in *Selected processes*.

A	B	L	M	N
List of Processes	Process ID	Human health [min. lost/Unit]	Human health [min. lost/FU]	fraction human health
Manufacturing (Materials + Processing)				
steel	21143	2.08	8.81E-03	-1.1%
Metal working	14178	2.28	9.68E-03	-1.2%
aluminium	502	3.48	1.18E-03	-0.1%
Copper	20335	2.19	2.80E-03	-0.3%
Extruded thermoformed plastic sheets	8749	0.63	4.57E-04	-0.1%
Polypropylene & polyurethane	17407	0.43	3.30E-04	0.0%
synthetic rubber	21656	2.13	5.05E-04	-0.1%
glas	9020	0.65	1.41E-04	0.0%
battery production, Li-ion, LiMn2O4, rechargeable	1090	18.01	3.84E-02	-4.8%
electronics, for control units	8391	29.94	3.10E-03	-0.4%
ethylene, paints, fluids	8639	0.27	2.70E-05	0.0%
tap water	21711	0.00	4.32E-06	0.0%
heat, district or industrial, natural gas	10352	0.00	3.93E-05	0.0%
electricity, photovoltaic, 570kWp open ground instal	7220	0.07	7.28E-04	-0.1%
Transport				
transport, freight train	22465	0.02	8.56E-05	0.0%
transport, freight, lorry 16-32 metric ton	22337	0.06	2.54E-05	0.0%
transport, freight, sea, container ship	22445	0.01	7.46E-04	-0.1%
Use phase				
Health benefit from driving phys. Activity	100002	-0.90	-9.01E-01	112.2%
Electricity Europe average	7666	0.12	3.16E-02	-3.9%
Disposal				
plastic mixture incineration	23993	0.09	1.53E-04	0.0%
Glass to landfill	23563	0.00	7.96E-07	0.0%
Avoided electricity plastic incineration	7666	0.12	-6.19E-04	0.1%
		Total min. lost/FU	-0.80	

Figure 12: Calculation tool showing heath considerations resulting in *Total min. lost/FU*.

5 Calculating Life Cycle Costing

Life Cycle Costing (LCC) estimates the total cost associated with the production, use, and end-of-life stages of a product or service. After defining the carbon footprint of your case study

using the *Carbon Footprint tool*, you can link the elementary flows of the selected process to the *Life Cycle Costing tool*.

The functional unit (FU) and the reference flows must be defined so that the corresponding cells in the tool can be completed. Any missing data should be added by searching the relevant Excel sheets or consulting reliable online sources. Enter the cost per unit (Cost/Unit) in column F; the tool will then automatically calculate the Cost per FU and the fraction costs. The data used for generating the graphs will populate automatically once all inventories in the tool have been fully completed.

6 Multicriteria decision making - Scenario Ranking

6.1 Analytical Hierarchy Process

Upon completion of the *Carbon footprint tool* and *Life cycle costing tool*, the various scenarios will be displayed in the first table of the *Scenario ranking* Excel sheet. The purpose of this sheet is to evaluate and rank these scenarios based on different sustainability aspects (economic, environmental, and social), utilizing weights that should be specified by the user. The final ranking of the scenarios will be visible in the *Total Score* column, which ranks scenarios from 0% to 100%, using different colors (the greener the better the ranking). Once the scenario table is completed (Columns A to D-E, Rows 1 to 6-15), the user should specify the weights of the various sustainability factors within the *Inputs* table (Figure 4). The user must determine these weights by modifying the *green, bold* values of the ratios (e.g., e-LCA/LCC, e-LCA/s-LCA). By doing so, the *Inputs* table will be automatically updated to reflect the calculations of the reverse negative, normalized values, and weighted values (Figure 5).

	F	G	H
Inputs			
e-LCA/LCC		8	
Consistency Ratio=		0.0	Must be <0.1

Figure 4: Example table for the weights that should be given for the AHP analysis

6.2 Weight ratio consistency

When more than two factors are evaluated using the AHP method, a consistency check should be performed to assess the robustness of the resulting decisions. The outcome of this check appears in the *Inputs* table (Figure 4) in the *Consistency Ratio* cell, shown in *green* colour, if the weight ratios are consistent and in *red* colour if they are not.

AHP		
Reverse Negative		
Petrol Vehicle	0.03	0.00
Electric Vehicle	0.00	0.09
Electric Vehicle - Wind	0.19	0.09
Electric Vehicle - Photovoltaic	0.28	0.09
Electric Vehicle - hard coal	0.26	0.09
Normalization		
Petrol Vehicle	0.12	0.00
Electric Vehicle	0.00	1.00
Electric Vehicle - Wind	0.67	1.00
Electric Vehicle - Photovoltaic	1.00	1.00
Electric Vehicle - hard coal	0.94	1.00
Weights		
	0.89	0.11
Petrol Vehicle	0.11	0.00
Electric Vehicle	0.00	0.11
Electric Vehicle - Wind	0.60	0.11
Electric Vehicle - Photovoltaic	0.89	0.11
Electric Vehicle - hard coal	0.83	0.11

Figure 5: Example AHP table

The pairwise comparison matrix forms the initial stage of the analysis. All missing cells must be filled in order to translate the specified weight ratios into weights. It is worth noting that the diagonal elements of the matrix must be equal to one. The standardized matrix is then constructed by normalizing the pairwise comparison matrix by the sum of each column. The final weights are obtained by calculating the average value of each row in the standardized matrix (Figure 6). The *consistency index*, *CI* must be provided by the user when more than two factors are considered. Finally, the *consistency ratio* has to be less than 0.1 in order to ensure robust scenario ranking analysis.

CI								
Pairwise comparison	e-LCA	LCC						
e-LCA	1	8						
LCC	0.125	1						
Sum	1.125	9						
Standardized Matrix	e-LCA	LCC	Weight					
e-LCA	0.89	0.89	0.89					
LCC	0.11	0.11	0.11					
CI and CR worksheet	e-LCA	LCC	Weighted sum value	Weighted sum value/weight	λ_{max}	CI	CR	must be
e-LCA	0.89	0.89	1.78	2	2	0	0	<0.1
LCC	0.11	0.11	0.22	2				

Figure 6: Example CI table

7 Absolute Sustainability

Absolute environmental sustainability assessment (AESA) aims to quantify if a practice is sustainable in the true meaning of the word, i.e. a practice that can be sustained indefinitely without serious/ significant consequences. Unlike relative sustainability assessments, which simply compare technologies to identify which is *more sustainable*, AESA seeks to evaluate whether a technology is *sustainable enough* to operate within the planet's ecological limits.

7.1 Planetary Boundaries

There are multiple planetary boundaries for different kinds of emissions (for instance, land use, or acidification), but this tool focuses only on climate change. Hence, to perform the AESA, a planetary boundary in terms of CO₂-eq emissions is needed.

The absolute sustainability calculations can be found in the 'absolute sustainability' sheet. In Fig. 7, the calculation of the PB per person in the tool can be found, assuming the remaining budget is evenly distributed across the world population (i.e. egalitarian approach). The global planetary boundary will be given in cell C6, with the source in D6. The world population is given in cell C7, and the consequent PB per person is given in cell C10. In cells C8, the three-letter country code for the considered country should be inputted, and C9 is only a conversion multiplier.

	A	B	C	D
3	Determination of Absolute Sustainability: Climate Change			
4				
5		Unit	Values	Source
6	Planetary Boundary (PB) - climate change in CO ₂ eq	kgCO ₂ eq/yr	6.81E+12	Björn et al. 2015
7	World population on 15.11.2022	pers	8.00E+09	
8	Country (see ISO code in "World carbon footprint" tab)	Denmark	DNK	
9	Transformation multiplier	t/kg, kg/g...	1000	
10	Planetary Boundary (PB) - CO ₂ eq impacts per person	kgCO ₂ eq/pers/yr	851.5	

Figure 7: Using an egalitarian approach to calculate the planetary boundary per person

Then, there are various ways in which this personal 'budget' can be allocated across all the various products and services each person uses in a year, private or public. This tool makes use of two different methods, grandfathering (GF) and final consumption expenditure (FCE).

7.2 Allocation Methods

7.2.1 Final Consumption Expenditure

Final Consumption Expenditure is based on the idea of using the purchase value of a product to determine its allocated carbon emissions from the carbon budget. Firstly, we determine the percentage of a person's budget spent on the activity in question. We then assume that the same proportion of the person's carbon budget should be spent on this activity. This allows us to calculate the sustainable boundary assigned to this specific activity. As we cannot evaluate

each person's budget individually, we assume the average GDP per capita to identify the total spendings of an average person.

Excel Instructions and Example

1. In cell C9, the three-letter ISO code of the relevant country was added. (this can be found in the sheet 'world carbon footprint data'). The GDP per person of this country will then be added automatically in cell C14, in the local currency.
2. Add the total life cycle cost per FU in cell C15.
3. In cell C16, C15 is divided by C14, finding the share of the budget allocated to 1 FU.
4. In cell C17, the share of the sustainable boundary applicable to 1 FU is found, by multiplying C16 by C10, or the planetary boundary applicable to 1 person.

	A	B	C
13	Economic Allocation (FCE)		Reference System (Gasoline car)
14	GDP at purchasing power parity (PPP) per capita	CUR/pers/yr	361868
15	Life cycle cost per FU	CUR/FU	3.36
16	Share of total boundary attributable to 1 FU		9.27E-06
17	Sustainable boundary assigned to 1 FU	kgCO2e/FU	0.00790

Figure 8: Example of FCE

7.2.2 Grandfathering

Grandfathering assumes that the carbon footprint of each technology needs to reduce equally to meet the planetary boundary, and compares whether different technologies offer sufficient improvements compared with a reference scenario. Just like in the FCE case, we take as a starting point the average allowed emission per capita to stay within the planetary boundary. We then compare this to the actual emissions per capita in a country, thus finding the required reduction factor for an average person in this country. If we assume all technologies need to reduce their emissions equally (which is the basic assumption in GF), we have also determined how much each technology needs to reduce.

To find a sustainable boundary (in CO₂-eq), we need to compare with a reference scenario. By reducing the emissions of the reference scenario by the reduction factor, we find the allowable emissions per FU.

Excel Instructions and Example

1. In cell C20, you can find the current consumption of one person from the selected country.
2. In cell C21, calculate the reduction factor required to stay within the planetary boundary, by dividing the actual carbon use by the total budget per person (C20/C10)

19	Grand Fathering (GF)		Reference System (Gasoline car)
20	Total present consumption-based impact	kgCO2e/pers/yr	10287.36
21	Reduction factor to reach PB	-	12.08
22	Carbon footprint reference scenario	kgCO2e/FU	0.31
23	Sustainable boundary assigned to 1 FU	kgCO2e/V-km	0.0259

Figure 9: Grandfathering example

3. In cell C22, give the emissions of the reference scenario, in CO₂-eq, by taking it from the 'carbon calculation' sheet.
4. In cell C23, find the sustainable boundary per FU by dividing the reference scenario by the reduction factor (C22/C21).

7.3 Comparison

In rows 25-30, we can now compare the absolute sustainability of each of the scenarios. In row 28, the calculated carbon impact for each of the scenarios should be added, by taking them from the 'carbon calculation' tab. Rows 29 and 30 should then be filled in automatically, and display the required factor reduction for the technology to be considered absolutely sustainable. If this factor is less than 1, the technology can be considered absolutely sustainable according to the respective allocation method.

	A	B	C	D	E	F	G	H	I
25	Conclusion Impact								
26									
27			Reference System (Gasoline car)	EV - Hard coal	EV - EU mix	EV - Wind	EV - Solar	E-Bike - EU mix	Bike
28	Carbon footprint impact per FU	kgCO2eq/FU	0.31273	0.34650	0.15886	0.06756	0.08457	0.01489	0.00762
29	Absolute Sust. Ratio & exceedance FCE		39.6	43.9	20.1	8.6	10.7	1.9	1.0
30	Absolute Sust. Ratio & exceedance GF		12.1	13.4	6.1	2.6	3.3	0.6	0.3

Figure 10: Absolute sustainability of various scenarios